

Basic Principles of Medicine 2

Module: Endocrine & Reproduction

ER 23: Nutritional Aspect of Pregnancy, Lactation and Infant Nutrition

Dr. Felicia Ikolo
fikolo@sgu.edu
St Georges University ©

Acknowledgement
Dr. Margit Trotz

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Objectives: DLA: Nutritional Aspects of Lactation and Infant Nutrition

- DLA to be reviewed before “Nutritional Aspects of Pregnancy, Lactation and Infant Nutrition”

SOM.MK.III.BPM2.1.ER.2.BCHM.1055	Highlight the major differences in the composition of colostrum, transition milk and mature milk during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1056	Discuss the advantages of human milk compared to animal milk and infant formula.
SOM.MK.III.BPM2.1.ER.2.BCHM.1057	Review WHO guidelines about breastfeeding and HIV.
SOM.MK.III.BPM2.1.ER.2.BCHM.1058	Identify the age range of neonate and infant. Discuss Vitamin K deficiency bleeding (VKDB) in the neonate and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1059	Identify micronutrients that are low in human breast milk. Highlight the dietary modifications (iron) to be implemented after 6 months of age.
SOM.MK.III.BPM2.1.ER.2.BCHM.1060	Describe bioavailability of iron and factors that reduce inorganic iron uptake.

Objectives: Nutritional Aspects of Pregnancy, Lactation and Infant Nutrition

SOM.MK.III.BPM2.1.ER.2.BCHM.1062	List the consequences of maternal malnutrition on: Maternal health, Fetal health and the infant during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1063	Review the need for nutritional balance and the consequences of deficiencies on mother, fetus and infant as it relates to: Energy requirements and weight gain (Including carbohydrates and Omega 3 fatty acids like DHA), Proteins (Including enzymes and immunoglobulins in breast milk), Iron (Microcytic iron deficiency anemia), Folate (Macrocytic anemia and neural tube defects), Calcium, Vitamin A, Zinc and Iodine (congenital hypothyroidism).
SOM.MK.III.BPM2.1.ER.2.BCHM.1064	Explain the importance of safe handling and preparation of food, especially during pregnancy and lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1065	Discuss the effects and list some compounds that may pose health risks during pregnancy and lactation like Caffeine, Smoking, Alcohol (fetal alcohol syndrome), Drugs: medical and non-medical, Toxins and teratogens.
SOM.MK.III.BPM2.1.ER.2.BCHM.1066	Discuss the effect of alcohol on the fetus and newborn including fetal alcohol syndrome.
SOM.MK.III.BPM2.1.ER.2.BCHM.1067	Review the significance of screening for gestational diabetes in the expectant mother and explain the complications of maternal hyperglycemia on fetal development including fetal macrosomia.
SOM.MK.III.BPM2.1.ER.2.BCHM.1068	Evaluate the need to monitor weight of the expectant mother, infant and the lactating mother and provide counseling to achieve pre-pregnancy weight.
SOM.MK.III.BPM2.1.ER.2.BCHM.1069	List specific macronutrients required for neonate and infant growth.
SOM.MK.III.BPM2.1.ER.2.BCHM.1070	Discuss lactose content in human breast milk. Explain significance of milk exclusion in congenital lactase deficiency and in classical galactosemia.
SOM.MK.III.BPM2.1.ER.2.BCHM.1071	Describe the function of human milk oligosaccharides (HMOs) regarding the GI tract and immune system in infants. Discuss the change of serum levels of immunoglobulins in the fetus and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1072	Discuss the importance of DHA and arachidonic acid in breast milk related to brain growth and eye development
SOM.MK.III.BPM2.1.ER.2.BCHM.1073	Compare and contrast composition of cow's milk and breast milk.
SOM.MK.III.BPM2.1.ER.2.BCHM.1074	Highlight the significance of dietary essential amino acids and the conditionally essential amino acids needed in infants (positive nitrogen balance).

Recommended Readings

- ✓ It is recommended that you review and study the **DLA** before coming to this lecture.
- ✓ **Lippincott Illustrated Reviews, Biochemistry, 9th Edition:**
 - 🔗 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - ✓ Chapter 27: Nutrition: Overview and Macronutrients
 - ✓ Chapter 28: Micronutrients: Vitamins
 - ✓ Chapter 29: Micronutrients: Minerals
- ✓ **Other references** - This lecture was compiled with relevant content from different publications:
 - ✓ **Review: Maternal Diet and Nutrient Requirements in Pregnancy and Breastfeeding.** An Italian consensus Document Marangoni et al, Nutrients 2016, 8, 629 (17 pages). <https://doi.org/10.3390/nu8100629>
 - ✓ **Calcium metabolism during pregnancy and lactation.** De Groot et al. NCBI bookshelf Canada 2015 (25 pages) <https://www.ncbi.nlm.nih.gov/books/NBK279173>
 - ✓ **Pharmacokinetics of ethanol in the maternal-fetal unit.** Zeiner and Koren, J Popul Ther Clin Pharmacol 2013, 20 (3) e259-265. <https://www.ncbi.nlm.nih.gov/pubmed/24142064>
 - ✓ Diab L.K., & Haemer M.A., & Primak L.E., & Krebs N.F. (2022). **Normal childhood nutrition & its disorders.** Bunik M, & Hay W.W., & Levin M.J., & Abzug M.J.(Eds.), Current Diagnosis & Treatment: Pediatrics, 26e. McGraw-Hill Education. <https://accessmedicine.mhmedical.com/content.aspx?bookid=3163§ionid=266216083>
 - ✓ Berthold Koletzko, K.M. Godfrey, Lucilla Poston, Hania Szajewska, Johannes B. van Goudoever, Marita de Waard, Brigitte Brands, Rosalie M. Grivell, Andrea R. Deussen, Jodie M. Dodd, Bernadeta Patro-Golab, Bartlomiej M. Zalewski, EarlyNutrition Project Systematic Review Group; **Nutrition During Pregnancy, Lactation and Early Childhood and its Implications for Maternal and Long-Term Child Health: The Early Nutrition Project Recommendations.** *Ann Nutr Metab* 28 February 2019; 74 (2): 93–106. <https://doi.org/10.1159/000496471>

Healthy nutrition during pregnancy and lactation benefits the mother, fetus and infant.



Consequences of Maternal Malnutrition

1. Maternal health

Increased risk of **maternal complications**, weakness, infections and death. Deficiency of iron can lead to **microcytic anemia** and deficiency of folates can lead to **macrocytic anemia**.

2. Fetal health

Increased risk of **growth retardation**, low birth weight, and prematurity. Occurrence of birth defects like **NTD**, including brain damage and **congenital hypothyroidism**, in severe cases **fetal mortality**.

3. Infant during lactation

Increased risk of growth retardation, negative nitrogen balance and insufficient immune system leading to recurrent **respiratory infections**.

Recommended Intake of **Macronutrients** During Pregnancy and Lactation

1. Protein

Increased dietary intake up to **72 g/day** is recommended during **pregnancy** for **protein synthesis** for maternal tissues, placenta, fetal growth and during **lactation** for **milk** synthesis. **Maternal immunoglobulins** are provided to the fetus via blood during pregnancy and dietary to the infant in human milk.

Pregnancy: Positive nitrogen balance

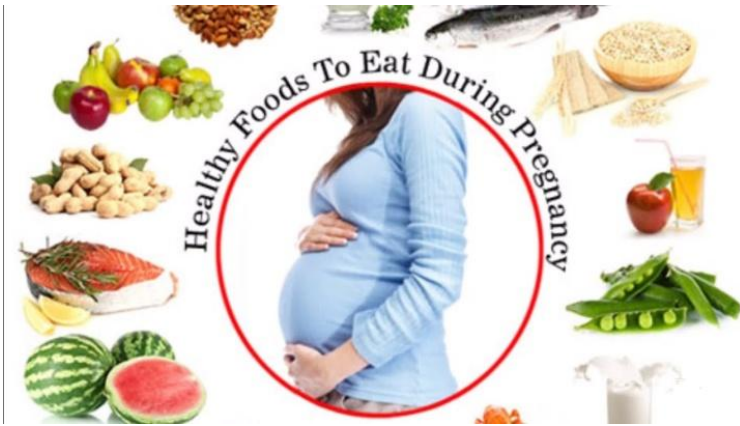
2. Fats

Increased maternal dietary intake of **DHA** during **pregnancy** and **lactation** is highly recommended as DHA is essential for **brain** and **vision development** in fetus and infant.

3. Carbohydrates

No increase of dietary **carbohydrates** is recommended as this could lead to **maternal overweight**.

Recommended Intake Of **Micronutrients** During Pregnancy And Lactation

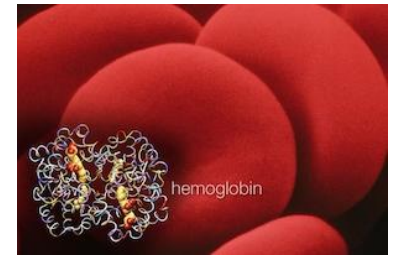


1. Iron
2. Folate and Zinc
3. Calcium
4. Vitamin A
5. Iodine





1. Iron is needed for RBC synthesis and fetal brain development



The **iron** requirement **increases dramatically in the 3rd trimester** and is mainly used for expansion of **maternal blood volume** and **fetal RBC** synthesis. Iron is needed for fetal brain development, myelination and neurotransmitters. The **growth** of the **fetal liver** is accelerated, and **iron is stored** to last up to about **6 month** after birth.

Pregnancy: iron supplementation is recommended to prevent **maternal microcytic anemia** which can also lead to **premature delivery** and **perinatal mortality**.

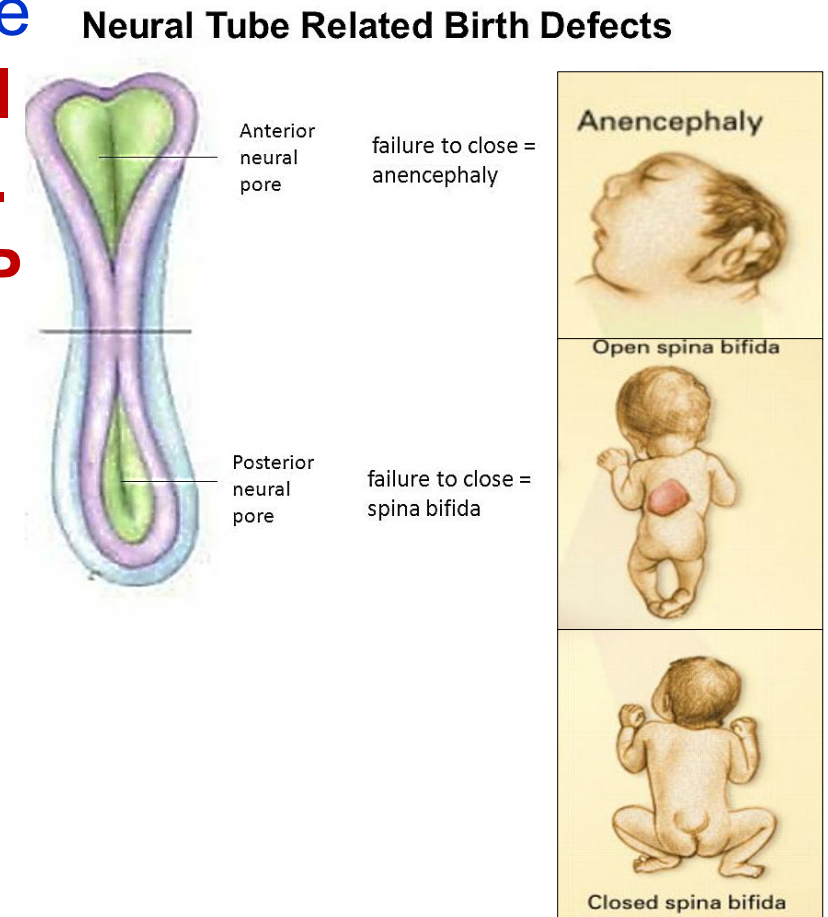
Lactation: the maternal iron demand is **normal** as **human milk is kept low in iron**. In addition, **human milk** contains **lactoferrin** which **binds iron** and **inhibits** growth of **iron-dependent bacteria** in the **GI tract** of neonates and infants.

2. Folates and Zinc are essential for cell division.



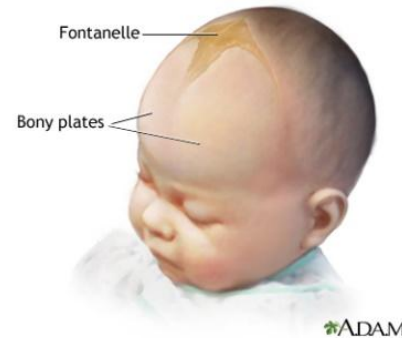
Deficiency of **folate** before conception or during the first trimester of pregnancy can increase risk of **fetal neural tube defects (NTD)** and to **maternal anemia**.
Fetal neural tube defect: **Elevated maternal serum AFP**

Increased **zinc** intake by about **50%** is needed to **prevent** NTD, abortion, stillbirth, preterm delivery, low birth weight (LBW) and small for gestational age (SGA).





3. Calcium is essential for bone and teeth growth in fetus and infant



Pregnancy and especially **lactation** are periods of **high calcium demand** for bone and teeth development. The **maternal RDA** for **calcium** intake of **1,200 mg/day** is however **not increased** for women with normal calcium levels.

The reason for this is that the **maternal calcium homeostasis changes** and leads to **more efficient absorption of dietary calcium** during **pregnancy** and **lactation**. This is **regulated** by **1,25 dihydroxy vitamin D (calcitriol)** and not by dietary calcium.

Lactation, however, leads to a temporary **mobilization** of **calcium** from **maternal bones**. This loss will be **refilled** after **weaning** and following **increased** maternal **estrogen** levels.



4. **Vitamin A** and retinoids are essential for growth and cell differentiation, vision and immune system.



5,000 IU

Pregnancy: Recommended increase in 2nd and 3rd trimester of the RDA (900 mcg/day) by **10%** to improve infant growth and development of skeleton, brain and lung.

Deficiency can lead to increased **fetal mortality** and **maternal night blindness**.

Teratogenicity: High intake of vitamin A or synthetic retinoids for treatment of **acne** or **psoriasis** during **pregnancy** can be **teratogenic** (more than 1,500 mcg) resulting in **cleft lip, cleft palate, hydrocephalus and heart malformations**.

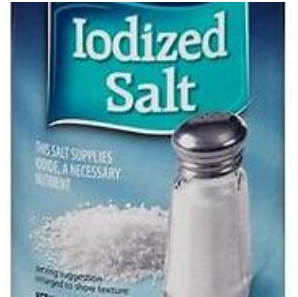


Lactation: Recommended increase to nearly twice of normal intake to ensure that the milk contains the necessary amount of retinoids for the infant.

Deficiency in infants can lead to **xerophthalmia, night blindness**

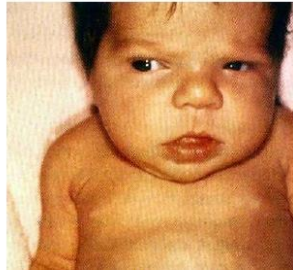


5. Iodine is essential for thyroid hormone synthesis for mother, fetus and infant.



Increased **iodine** intake by about **50%** is needed to **prevent** spontaneous abortion, perinatal mortality, birth defects and neurological disorders.

Congenital hypothyroidism:



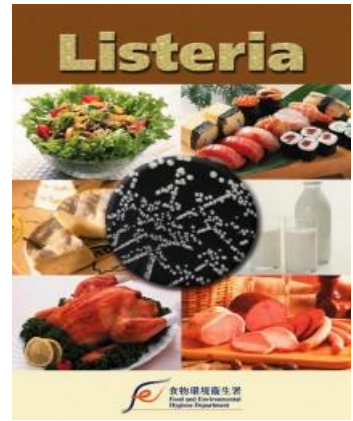
- ☐ Protruding tongue
- ☐ Growth retardation
- ☐ Jaundice
- ☐ Dry skin
- ☐ Slow reflexes
- ☐ Hoarse voice

Congenital hypothyroidism is characterized by **protruding tongue** (macroglossia), **hoarse cry**, **hypothermia**, **constipation**, **growth retardation** **intellectual disability**. Newborn screening for congenital hypothyroidism. Early detection and treatment to prevent intellectual disability.



Food safety especially during pregnancy and lactation:

Wash, pasteurize and cook well.



Toxoplasmosis and listeriosis are particularly dangerous during pregnancy.

Toxoplasmosis results from a single-cell human **parasite toxoplasma** carried by many **animals** especially **cats**. Pregnant women should **not clean** a cat **litter box**.

Toxoplasma is also found in **contaminated water**, **raw meat** and **unpasteurized milk** and other **food products**. It can lead to **blindness**, **hearing loss** and **brain damage** of the baby.

Listeriosis results from contaminated food by the bacterium **listeria** and in severe cases can lead to **miscarriage**, **premature labor** and **stillbirth**.

Health Risks During Pregnancy and Lactation

Avoid smoking

- a. Increased risk of **preterm delivery** and **low birth weight** of neonate.
- b. Increased incidence of **asthma** and **bronchospasm** in **infants** exposed to smoke even in the prenatal period.
- c. Smoking **reduces DHA** levels in maternal plasma and in milk and leads to reduced supply to fetus and infant.

Avoid alcohol (Detail discussed in following slides).

Avoid drugs including medical drugs (teratogenicity).

Avoid toxins like heavy metals and pesticides. High **mercury** content is found in some **fish**. **Pesticides** and **herbicides** are found in plant food.

Reduce caffeine to less than **1 cup** of coffee/day.

Discuss exclusion diets (vegetarian and vegan diets)

Exclusion diets devoid of any animal food including eggs and milk should be discussed regarding possible **deficiency** of **micronutrients**.

Fetal Alcohol Syndrome Symptom, Cause, Risk, Prevention

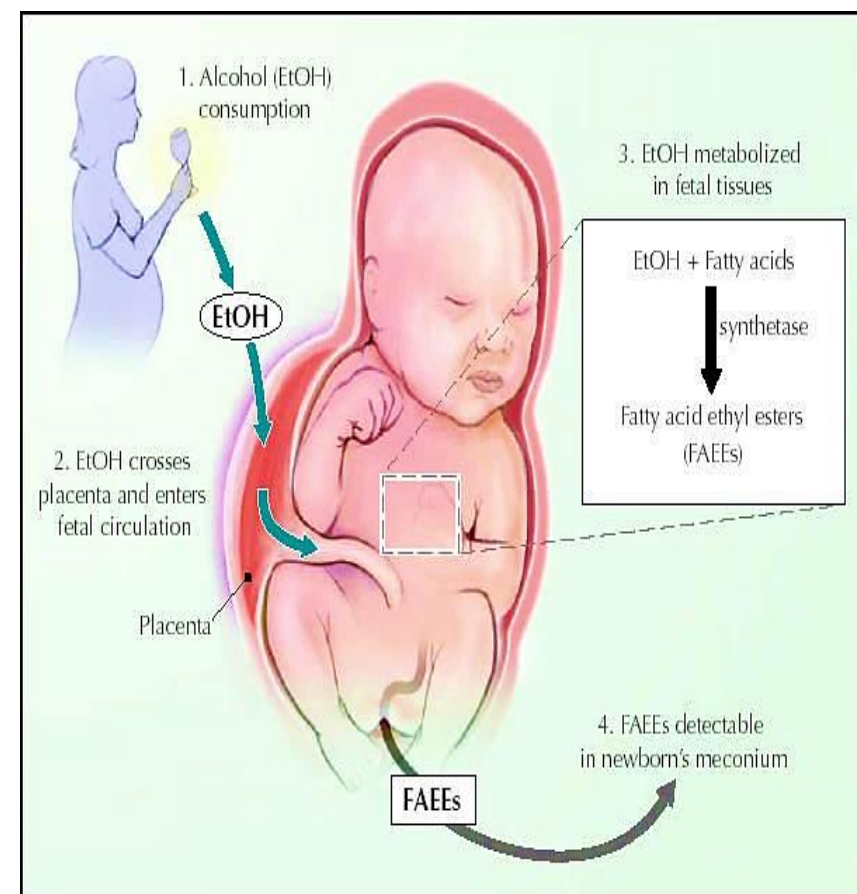


Parenting healthy babies
Homemade baby food recipes, health guides, baby care tips & more...

Fetal Alcohol Metabolism

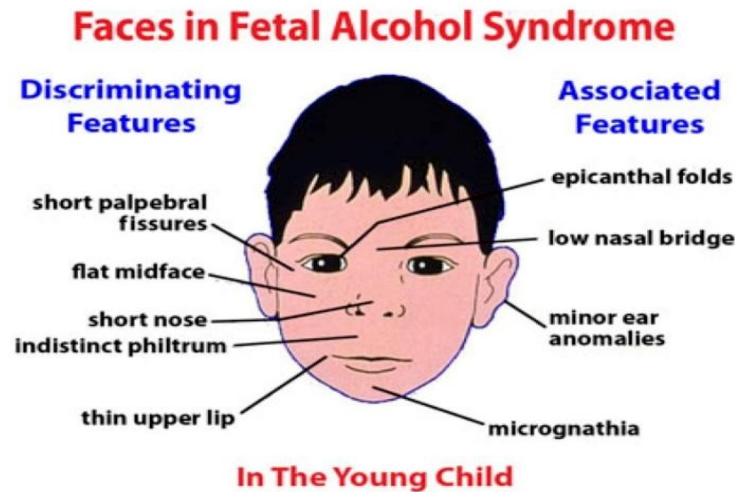


- **Alcohol** passes the placenta and leads to **high levels** in the **fetal circulation**.
- **Alcohol** is **differently metabolized** in the **fetus** when compared to **adults**. The adult liver uses ethanol to form acetaldehyde and then acetate. The synthesis of these hepatic enzymes increases only slowly during fetal development.
- In the **fetus**, **ethanol** is enzymatically **linked** to **fatty acids** or other molecules.
- **Fatty acid ethyl esters (FAEEs)** are **excreted** in **meconium** and can be measured as an **early marker** for possible **fetal damage** by ethanol.

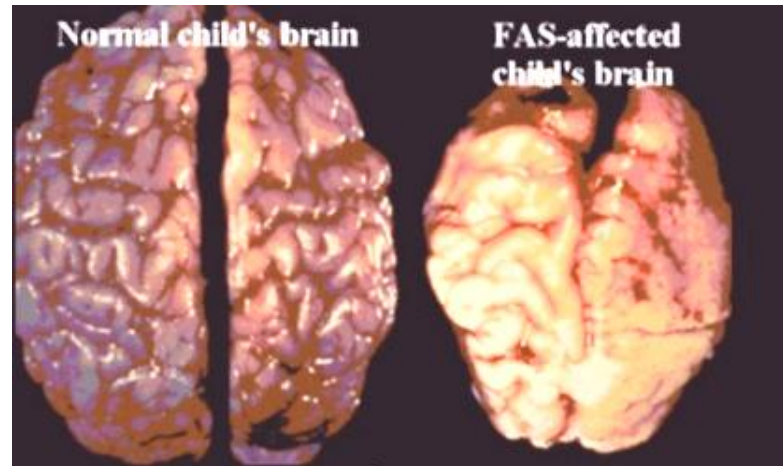


Fetal Alcohol Syndrome (FAS)

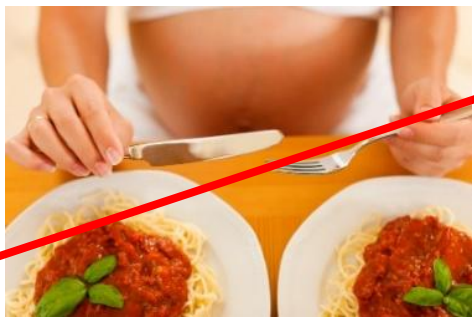
FAS can lead to **physical, cognitive, and behavioral** disabilities as it can **cause facial dysmorphology, intellectual disability**



Streissguth, 1994



Common: **facial abnormalities, low birth weight and low intelligence.**



Fetal Macrosomia

(Baby weighs more than 8 pounds)



Pregnant women do not need to eat for two. A **weight gain** of **25 to 35 lb** is recommended for women with a **BMI 18.5 - 24.9**.

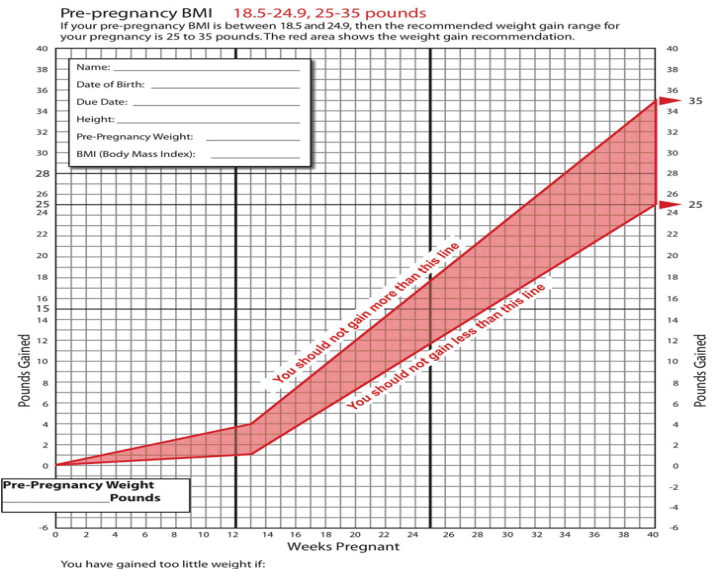
Lower weight gain recommended in overweight or obese women.

- **Monitor weight gain** and screen and control maternal **blood glucose levels**.
- Women with **gestational diabetes mellitus** and **overweight** women may deliver a “big baby” (fetal macrosomia) and other complications in newborn period.
- **Increased risk** of injury for mother and baby during delivery. A **caesarian section** is often recommended
- Females with gestational diabetes have an increased risk of **developing type 2** diabetes mellitus

Recommended intake of **calories** during pregnancy and lactation



Pregnancy Weight Gain Chart in Pounds



Pregnancy: Moderate step-wise increase in calories during the 1st, 2nd and 3rd trimesters (additional **70**, **260** and **500 kcal/day**).

Lactation: Additional **500 kcal/day** (first 6 months exclusive breastfeeding)

Growth charts of infants: Boys and girls

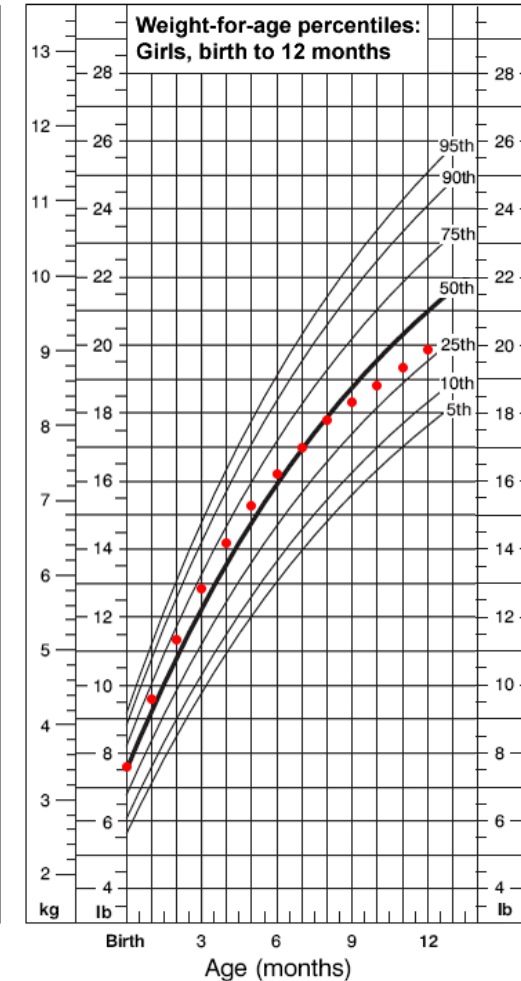
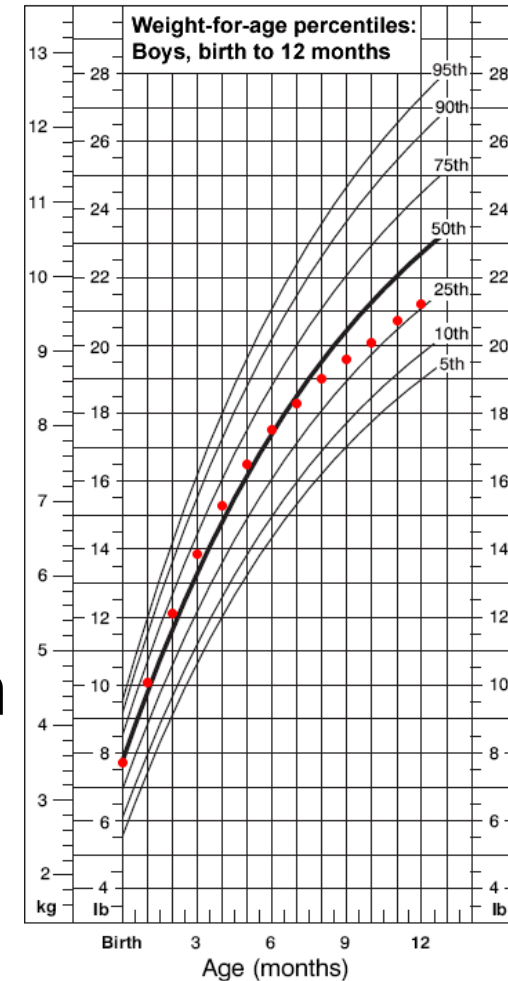
Positive nitrogen balance

- A. First 3 months: the baby may gain 2 lb per month.
- B. 6 months: the baby may gain 1 lb per month.
- C. 9 months: the baby may gain less than 1 lb per month.



Average Growth Patterns of Breastfed Infants

The red points plotted on the CDC Growth Charts represent the average weight-for-age for a small set of infant boys and girls who were breastfed for at least 12 months (see references).



Sources:

- Base chart -- CDC Growth Charts: United States, Published May 30, 2000.

Graphic by kellymom.com, 2004

- Breastfed baby data points -- WHO Working Group on Infant Growth. An Evaluation of Infant Growth: a summary of analyses performed in preparation for the WHO Expert Committee on Physical Status: the use and interpretation of anthropometry. (WHO/NUT/94.8). Geneva: World Health Organization, 1994, p.21.

Different stages of life after birth need adjusted nutrition



Premature less than 9 months pregnancy



Neonate up to 28 days old



Infants (babies) are up to 1-year-old

(Latin word *infans* means unable to speak)

Nutrition for neonates and infants requires a healthy diet at a time of intensive growth

Human milk or formula contains specific macronutrients:

1. **Carbohydrates**: Lactose (milk sugar)
2. **Fat**: Fatty acids of the $\omega 3$ and $\omega 6$ families.
3. **Proteins**: Whey and casein with essential amino acids.

Note: If **cow milk** is used to feed the infant, the diet is increased in proteins.
Cow milk is not recommended but often used, it **needs to be diluted** for better digestion.

1. Carbohydrates in human milk:

Large amount of **lactose** (milk sugar) which is cleaved in the small intestine to glucose and galactose by lactase.

	Lactose (g/L)	Fat (g/L)	Protein (g/L)
Human	7.2 (high)	4.1	1.3 (low)
Cow	4.6	3.9	3.3
Goat	4.4	3.7	3.1
Sheep	5.1	5.8	5.4

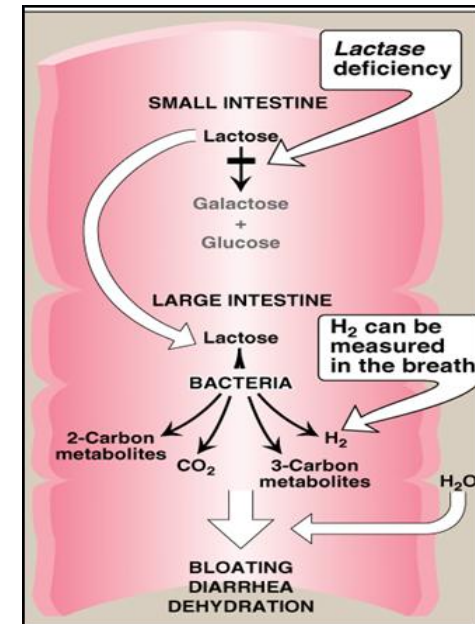
Congenital Lactase Deficiency: Stop breast feeding

Congenital Lactase Deficiency: Ingestion of milk (breast feeding) results in severe osmotic diarrhea and dehydration.

The **baby cries** due to the **painful bloating** which is often **mistaken for hunger**.

Feeding with milk (breast feeding) continues to cause pain and damage.

Management: **Lactose-free formula** and stop breast feeding



Classical Galactosemia: Stop breast feeding

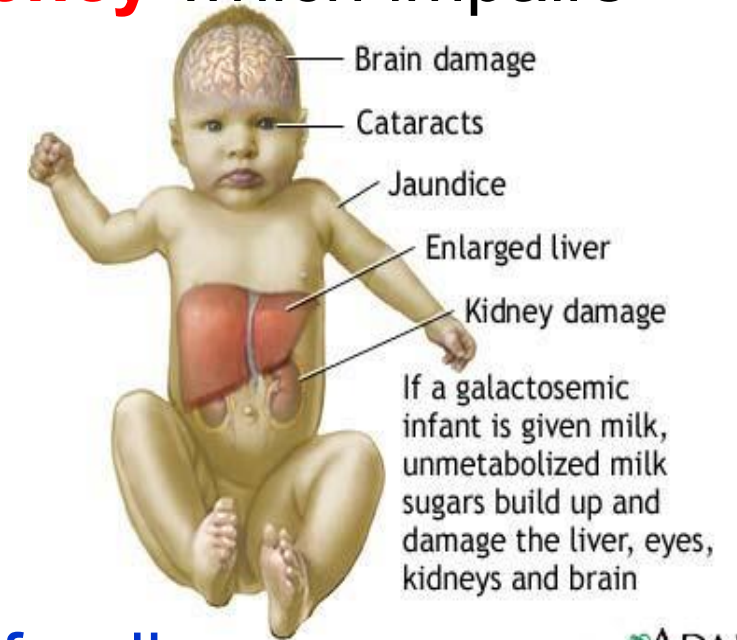
Newborn screening for galactosemia. Urine testing is positive for reducing sugars.

Classical Galactosemia: **Inherited enzyme deficiency** which impairs liver galactose metabolism

Characteristics of galactosemia:

- **Brain damage**
- **Cataracts**
- **Liver damage (jaundice and hepatomegaly)**
- **Neonatal sepsis**

Management: **Lactose-free formula** and stop breast feeding



ADAM.

Human milk stimulates the development of a healthy infant's gut and immune system

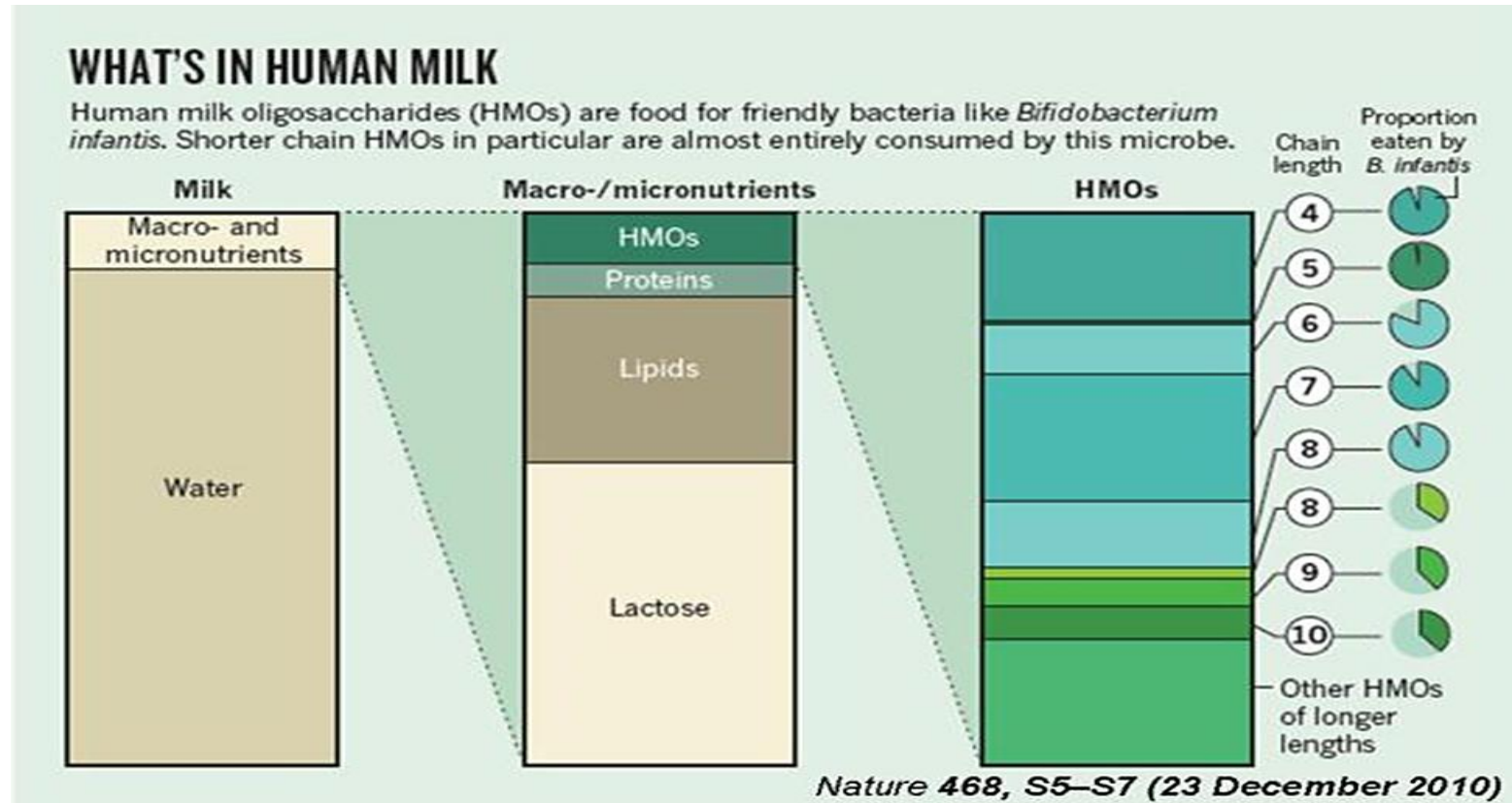
The GI tract **prior** to birth is **sterile** and the **immune system** is **immature**.

Human milk provides:

- a. **Human milk oligosaccharides (HMOs)** which are nutrients for “friendly” bacteria: *Bifidobacterium infantis* and *Lactobacillus acidophilus*.
- b. **The enzyme lysozyme** which protects the infant against *E. coli* and *Salmonella* and promotes the growth of healthy intestinal flora.
- c. **The immunoglobulin IgA** which protects the infant's intestine against pathogens.

Human milk oligosaccharides (HMOs) are food (prebiotics) for friendly bacteria (probiotics).

HMOs fermented by bacteria result in an **acidic pH** in the **gut** which makes the environment **less suitable** for **pathogenic bacteria**.



Serum levels of immunoglobulins in the fetus, neonate and infant

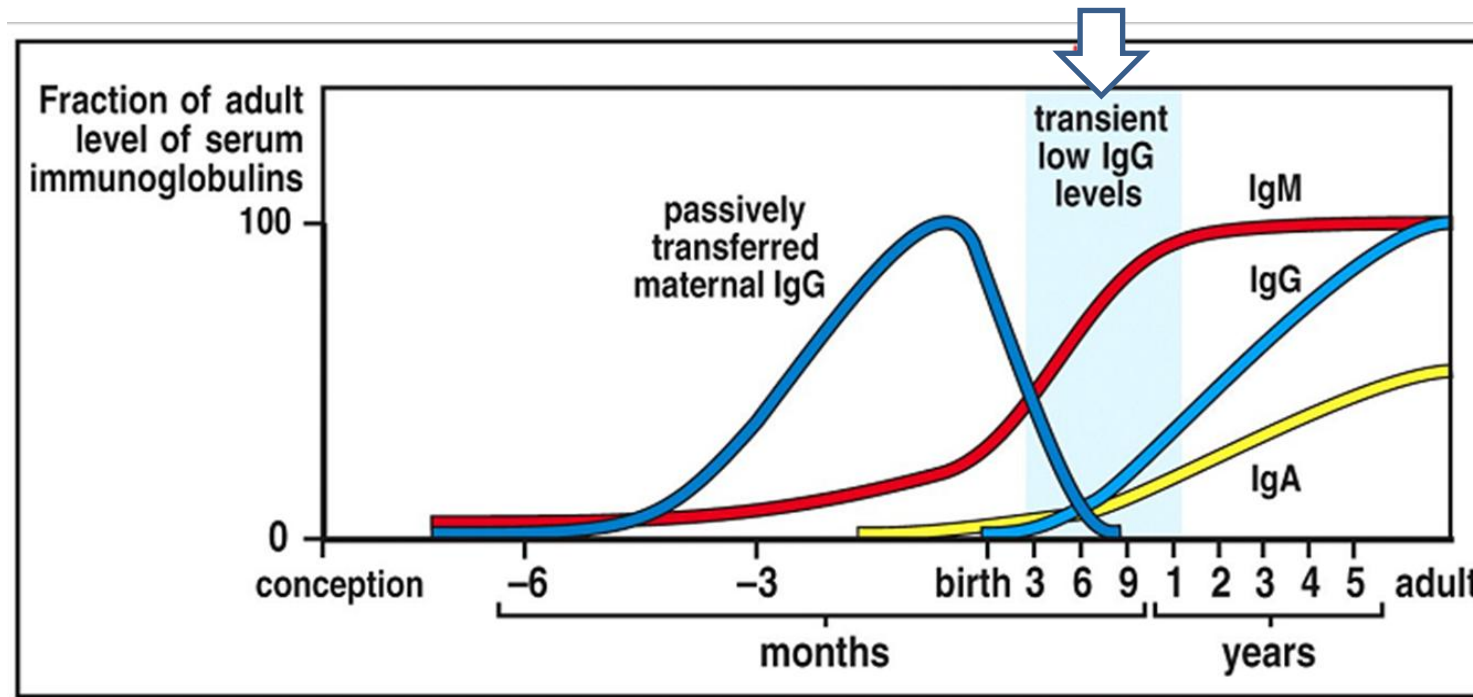
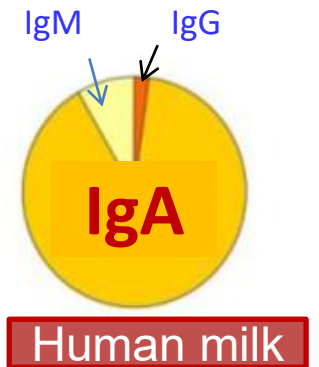


Figure 11-11 Immunobiology, 6/e. (© Garland Science 2005)

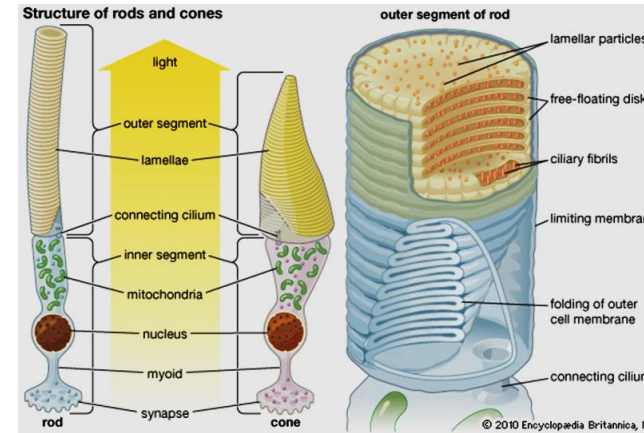
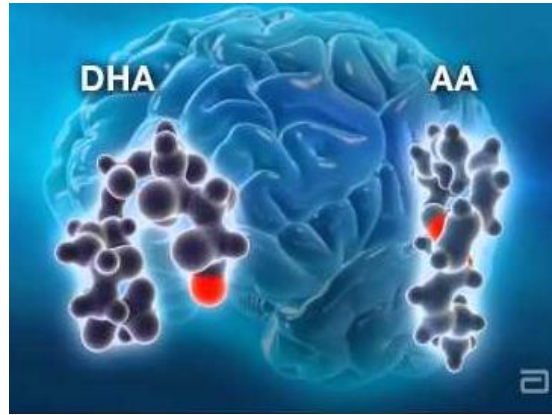
Passive transfer of **maternal IgG** to the fetus during pregnancy.
After birth, the infant starts **IgM** synthesis, followed by **IgG** and **IgA**.

Summary: Immunoglobulins

- **During pregnancy provided via the blood:**
 - **IgG** is the only antibody that **crosses** the **placenta** and is **passively** transferred to the **fetus**.
- **After birth synthesis of antibodies in infants:**
 - The newborn synthesizes **IgM** in the **early postnatal** period which is followed by **IgG** and **IgA**.
- **After birth provided in human milk:**
 - High levels of **IgA** (more than 90% of immunoglobulins) is found in human milk
 - **How can the protein IgA work when given via the diet?**
 - The partial digestion is slower for immunoglobulins than for other proteins. The antigen-binding sites remain mostly intact and provide protection from pathogens in the intestinal tract. **IgA** lines the immature gut.



2. Fats in human milk: DHA and arachidonic acid.

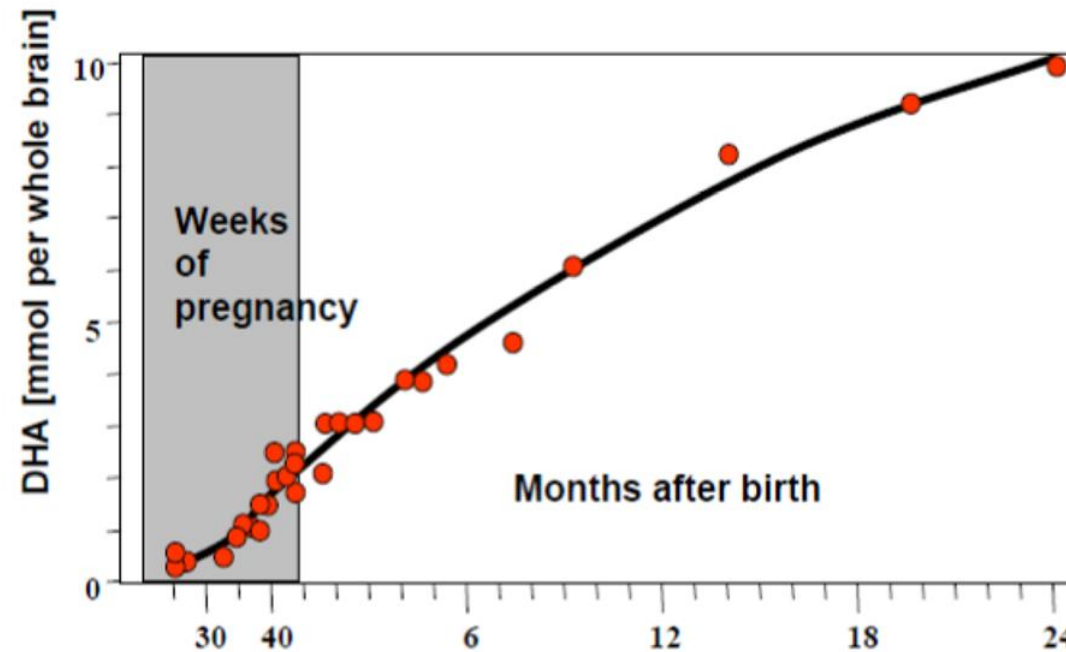


DHA ($\omega 3$) and arachidonic acid (AA, $\omega 6$) are able to cross the blood-brain barrier of the developing brain and are essential for **brain development** and **growth** and development in **vision**. These fatty acids lead to high fluidity of the membranes.

Note: A maternal **diet rich** in **DHA** increases **DHA** in human milk.

Brain growth and DHA

DHA is found in very **high concentration** in the **brain**.



DHA is taken up during **pregnancy** from the **blood** and after **birth** from human **milk**.

The brain triples in size in the first year of life

The head circumference is measured for normal development.

Measurement of head circumference



Infant brain growth



20 weeks



35 weeks



40 weeks

Human Milk compared to Cow milk and Infant Formula

Breast Milk vs Cow and Infant Formula Composition

Content per liter	Human	Cow	Infant Formulas		
			Preterm	Conv.	Milk-free
Energy (kcal)	690	660	676	676	676
Protein (g)	9	35	20	18	17
Taurine (mg)	80-40	1	48-57	40	
Fat (g)	45	37	35	35	35
Carbohydrates	68	49	74	79	75
Lactose	68	49	30	55	0

Whey/Casein ratio
of
80/20 – 60/40

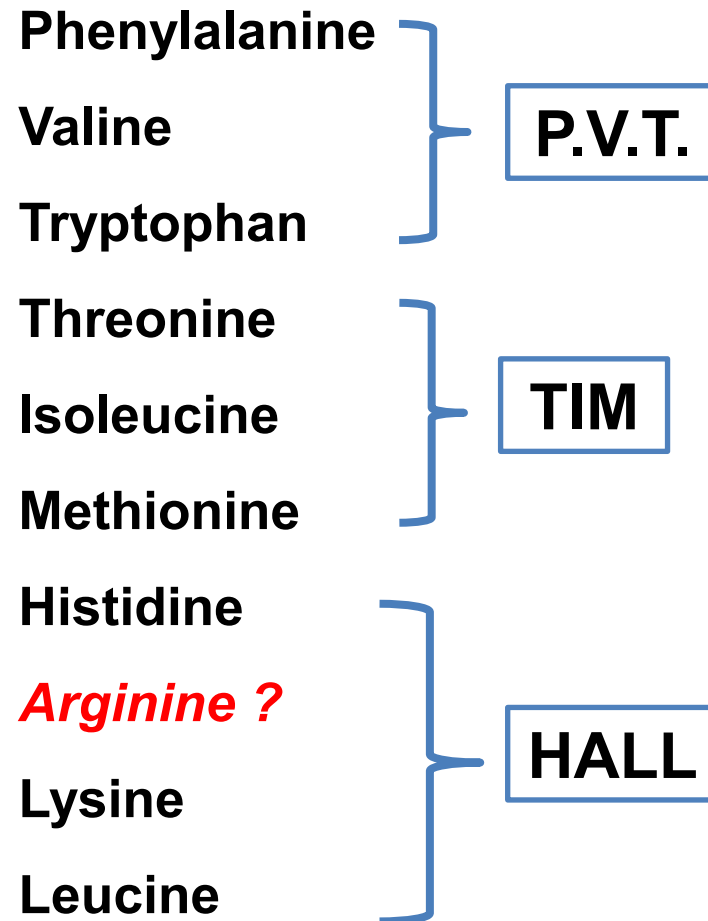
Whey/Casein ratio
of
20/80

Reminder: What Is The Advantage Of **Whey**?

- a. **α -lactalbumin:** **Predominant protein** in whey and is often referred to as **whey protein**. Rich in cysteine and tryptophan and easier to digest than casein.
- b. **Lactoferrin:** binds iron and **inhibits** growth of pathogenic **iron-dependent bacteria** in the **GI tract** of neonates and infants.
- c. **Lysozyme:** An **enzyme** that **protects** the infant against ***E. coli*** and ***Salmonella***. It **promotes** the growth of **healthy** intestinal **flora** and has **anti-inflammatory functions**.
- d. **Immunoglobulins:** mostly **IgA** which inhibits binding of pathogens, and protects the intestine.

3. Proteins in human milk:

Human milk has all dietary essential amino acids needed for a positive nitrogen balance (infants)



Additional Requirements For Proper Growth Of Infant:

larger dietary amounts of:

Glutamine (Nucleotide synthesis)

Cysteine (protein tertiary structure)

Glycine (Heme and collagen synthesis)

Proline (Collagen synthesis)

Tyrosine (Melanin, hormone synthesis)



With the increased need for food in growth after birth:

How does the stomach capacity grow and increase?

	Neonate	2 month old baby	1 year old infant	Adult
Stomach capacity	1.5 oz	2-3 oz	7-10 oz	1 quart
Size	Strawberry 	Small kiwi 	Orange 	Small cantaloupe 

Thank you!

Dr. Felicia Ikolo
fikolo@sgu.edu